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Jumlah masing masing komunitas

- Louvain = 3
- Label Propagation = 4
- Girvan-Newman = 4

```python
import networkx as nx
import matplotlib.pyplot as plt
import community.community_louvain as community_louvain
```

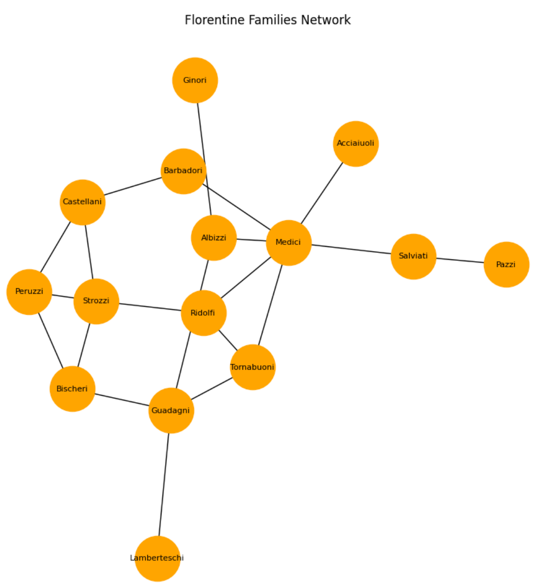
✓ 2.8s

```python
G = nx.florentine_families_graph()

print("Total node:", G.number_of_nodes())
print("Total edge:", G.number_of_edges())
```

✓ 0.0s

```
Total node: 15
Total edge: 20
```

```python
#visualisasi graph

plt.figure(figsize=(8,8))

pos = nx.spring_layout(G, seed=42)

nx.draw(
    G,
    pos,
    with_labels=True,
    node_size=2000,
    node_color="orange",
    font_size=8
)

plt.title("Florentine Families Network")
plt.show()
```

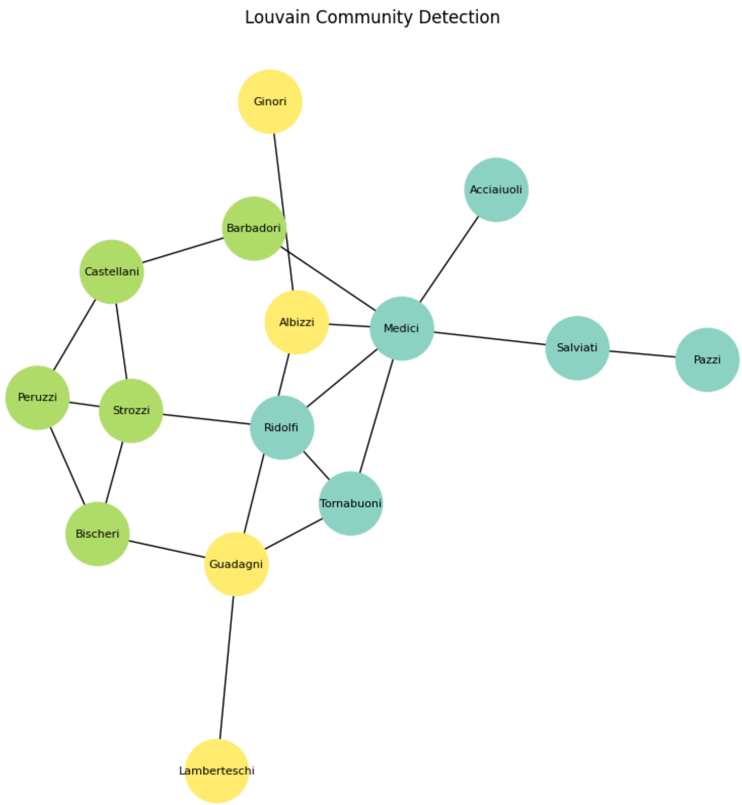


<< visualisasi graph

Louvain

```python
#louvain method
partition = community_louvain.best_partition(G)
values = [partition[node] for node in G.nodes()]

plt.figure(figsize=(8,8))

nx.draw(
    G,
    pos,
    with_labels=True,
    node_color=values,
    cmap=plt.cm.Set3,
    node_size=2000,
    font_size=8
)

plt.title("Louvain Community Detection")
plt.show()
```

✓ 0.1s




```python
# total community
total_komunitas = len(set(partition.values()))

print("Total komunitas:", total_komunitas)
```

✓ 0.0s

```
Total komunitas: 3
```

Propagation

```python
communities = nx.community.label_propagation_communities(G)
communities = list(communities)

print("Total komunitas:", len(communities))
```

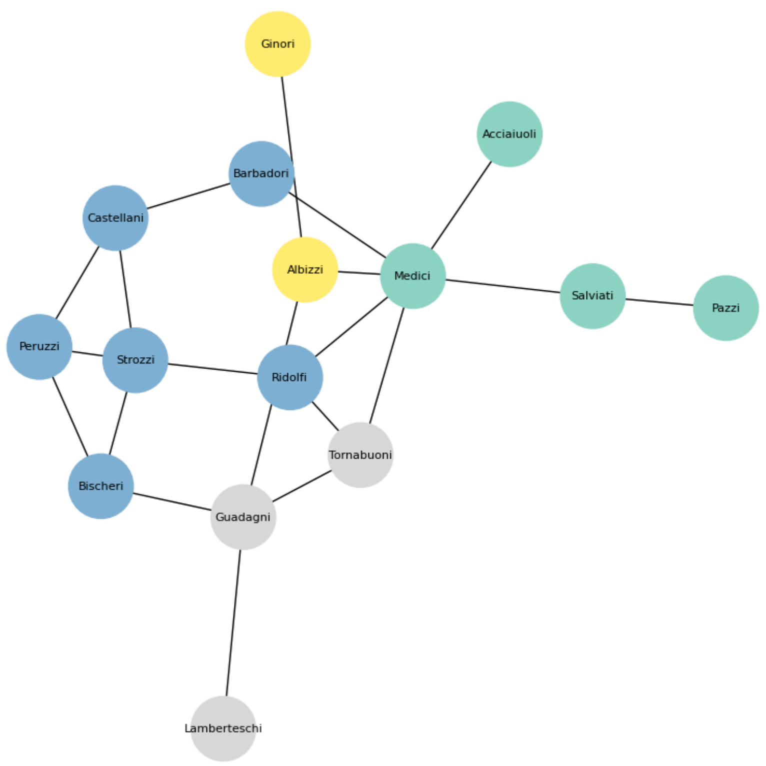
[11] ✓ 0.0s

```
... Total komunitas: 4
```

```python
# label propagation
node_colors = {}
for i, community in enumerate(communities):
    for node in community:
        node_colors[node] = i

colors = [node_colors[node] for node in G.nodes()]

plt.figure(figsize=(8,8))

nx.draw(
    G,
    pos,
    with_labels=True,
    node_color=colors,
    cmap=plt.cm.Set3,
    node_size=2000,
    font_size=8
)

plt.title("Label Propagation")
plt.show()
```

✓ 0.1s

Label Propagation



Girvan-Newman

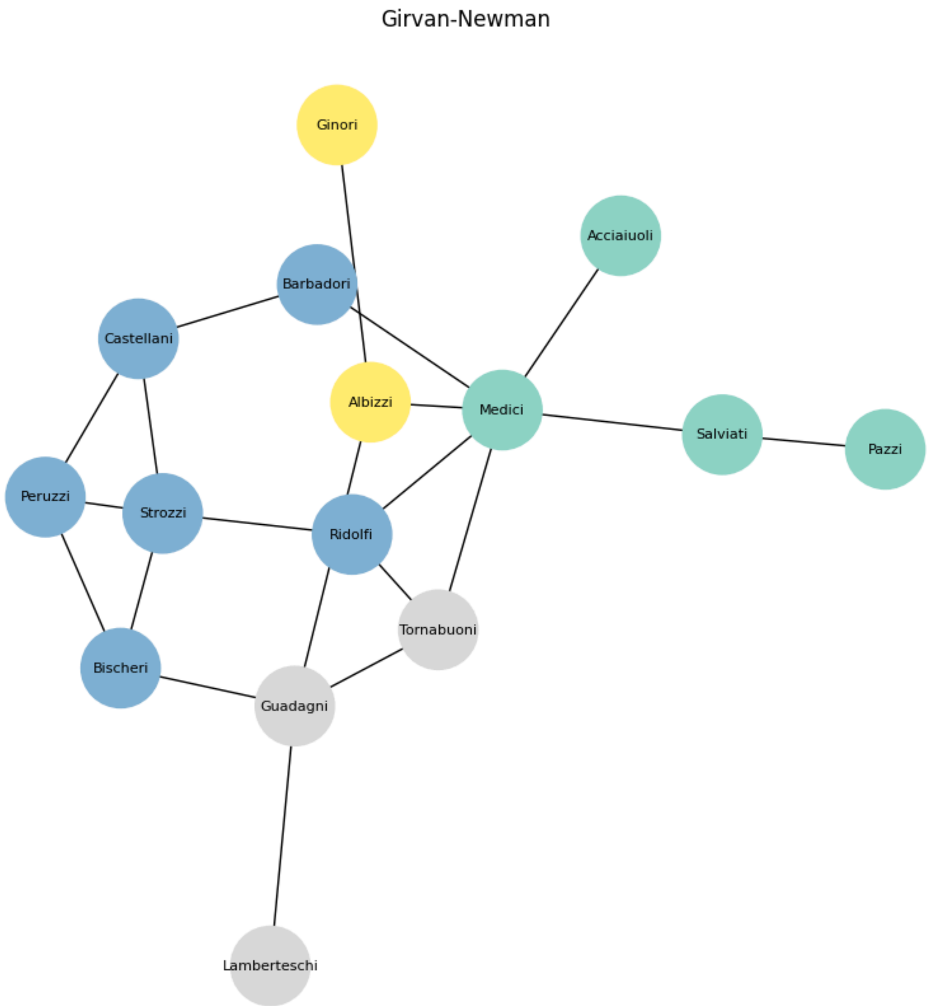
```python
communities = nx.community.label_propagation_communities(G)
communities = list(communities)

print("Total komunitas:", len(communities))
```

✓ 0.0s

```
Total komunitas: 4
```

```python
node_colors = {}
for i, community in enumerate(communities):
    for node in community:
        node_colors[node] = i

colors = [node_colors[node] for node in G.nodes()]

plt.figure(figsize=(8,8))

nx.draw(
    G,
    pos,
    with_labels=True,
    node_color=colors,
    cmap=plt.cm.Set3,
    node_size=2000,
    font_size=8
)

plt.title("Girvan-Newman")
plt.show()
```

✓ 0.1s



analisis :
Algoritma Louvain menghasilkan jumlah komunitas yang lebih sedikit karena berfokus pada pembentukan komunitas yang besar dan stabil. Sementara itu, Label Propagation dan Girvan-Newman menghasilkan komunitas yang lebih banyak karena lebih sensitif terhadap hubungan antar node. Secara keseluruhan, hasil pembagian komunitas pada dataset *Florentine Families* sudah sesuai dengan konteks data. Keluarga yang memiliki hubungan dekat, seperti hubungan politik atau pernikahan, cenderung berada dalam komunitas yang sama. Hal ini menunjukkan bahwa jaringan memiliki struktur komunitas yang cukup jelas dan setiap algoritma memiliki cara kerja yang berbeda dalam mendeteksi komunitas.